

Genomic Repeat inference from Depth (GRiD): A Pipeline for Haplotype-Resolved Large VNTR Copy Number Estimation

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Summary

Variable number tandem repeats (VNTRs) are genomic regions in which a sequence motif is repeated in tandem a variable number of times across individuals. Large VNTRs, repeat units spanning hundreds to thousands of base pairs, are a substantial source of structural variation in the human genome and have been linked to complex traits and disease (Mukamel et al., 2021). Because repeat units in these VNTRs can be more than 35 times the length of the standard 150-bp short sequencing reads, large VNTRs cannot be genotyped by conventional variant-calling approaches. Furthermore, accurate copy number estimation from short-read whole-genome sequencing (WGS) remains challenging. A well-characterized example is the Kringle IV type-2 (KIV-2) VNTR within the LPA gene, which is the primary genetic determinant of lipoprotein(a) [Lp(a)] concentration (Mukamel et al., 2021; Utermann et al., 1987). Lp(a) is a causal and highly heritable ($h^2 > 70\%$) atherosclerotic cardiovascular disease (ASCVD) risk factor elevated in an estimated 1.5 billion people globally (Reyes-Soffer et al., 2022). Both Lp(a) concentrations and effect of LPA genetic variants vary substantially across ancestry groups, underscoring the importance of ancestry-aware genetic analyses and risk prediction (Kronenberg et al., 2022; Nordestgaard et al., 2010; Tsimikas, 2017). The KIV-2 repeat unit spans ~5.5 kb, with individuals carrying 1–40 copies per haplotype, and KIV-2 copy number is inversely correlated with Lp(a) concentration (Clarke et al., 2009; Kamstrup et al., 2009; Mukamel et al., 2021; Utermann et al., 1987). Accurate estimation of KIV-2 copy number therefore has important implications for genetic studies of Lp(a) and cardiovascular disease, including analyses of ancestry-related variation and genetic risk (Kronenberg et al., 2022; Mukamel et al., 2021). While this VNTR serves as a region of great interest, prior work has been limited to custom pipelines, limiting applications by the broader field.

GRiD (Genomic Repeat inference from Depth) is an open-source Python pipeline that addresses this methodological gap. GRiD estimates diploid and haplotype-resolved VNTR copy number from short-read WGS data. GRiD combines read-depth normalization, read depth profile-matched nearest-neighbor estimation, and identity-by-descent (IBD)-based haplotype inference (Hujoel et al., 2026). Although GRiD was developed and validated for the LPA KIV-2 locus, its core framework is locus-agnostic and applicable to any large VNTR with sufficient sample size. The pipeline is designed to be accessible to researchers without extensive bioinformatics expertise.

Statement of need

Large VNTRs present a shared set of computational challenges that place them outside the scope of existing repeat genotyping software and standard genomic analysis pipelines. Their repeat units are orders of magnitude larger than the short tandem repeats (STRs) targeted by most available tools (Bakhtiari et al., 2018; Dolzhenko & others, 2019; Mousavi et al., 2019; Willems et al., 2017), and in individuals with high-copy-number alleles the VNTR region can span hundreds of kilobases, making read-pair spanning approaches difficult with standard paired-end reads. Haplotype resolution from short-read data can be important for downstream genomic analyses that depend allele-specific copy number, including analyses of statistical associations, genomic coverage, and haplotype based tract, particularly given the expense and difficulty of obtaining long-read data for large epidemiological cohorts. GRiD is designed to address these challenges directly through three features: (1) a read-depth normalization and nearest-neighbor framework that is locus-agnostic and parameterized entirely through a configuration file, (2) an IBD-based iterative phasing algorithm (Hujoel et al., 2026) that decomposes diploid copy number into haplotype-specific estimates using cohort relatedness, and (3) a single installable pipeline that encapsulates the full workflow from short-read WGS to haplotype-resolved copy number in an intuitive, reproducible, multi-threaded form.

Although several software packages address aspects of tandem repeat analysis, no existing open-source tool provides an end-to-end workflow for estimating both diploid and haplotype-resolved copy number from short-read whole-genome sequencing data at modern biobank scales. Consequently, many studies have relied on custom analysis pipelines, limiting reproducibility, portability, and cross-cohort compatibility. GRiD fills this gap by providing a configurable, documented, and openly available implementation of these methods.

State of the field

Several mature software packages exist for genotyping tandem repeats from short-read sequencing data. ExpansionHunter (Dolzhenko & others, 2019), and HipSTR (Willems et al., 2017) were developed primarily for short tandem repeats (STRs) whereas GangSTR (Mousavi et al., 2019) was developed for short VNTRs (defined by a repeat motif length of up to 20bp). However, these methods are designed for repeat lengths where short reads can still provide sufficient direct evidence about repeat structure. Likewise, TRTools (Mousavi et al., 2021) provides downstream analysis utilities for STR genotypes but does not extend to large VNTR copy number estimation. advNTR (Bakhtiari et al., 2018) supports genome-wide VNTR genotyping using hidden Markov models but is designed for repeat units that are substantially shorter (<150 bp) than loci such as the 5.6 kb LPA KIV-2 repeat. Earlier methods such as VNTRseek (Gelfand et al., 2014) detected shorter VNTRs by comparing read-derived repeat counts to a reference catalog, while GtTR (Ganesamoorthy et al., 2018) estimated absolute target VNTR copy number by scaling short-read depth across locus boundaries against a baseline genotype derived from long reads, though it remains unable to phase individual diploid alleles. More recently, danbing-tk (Lu et al., 2021) introduced repeat-pangenome graphs for genome-wide profiling of VNTR length and motif composition from short-read sequencing data. This approach represents VNTR sequence diversity using haplotype-resolved assemblies and estimates VNTR dosage from graph-specific k-mer counts. Recent advances in long-read sequencing have enabled direct characterization of complex repeat regions (Didericksen & others, 2024). For example, Kivvi (PacificBiosciences, 2026) estimates KIV-2 copy number from PacBio HiFi sequencing data. However, most large population cohorts (eg, 1000G (The 1000 Genomes Project Consortium, 2015), TOPMed (Taliun et al., 2021), and the Genotype-Tissue Expression (GTEx) project (GTEx Consortium, 2017)) remain based on short-read whole-genome sequencing, motivating methods that can estimate VNTR copy number without requiring long-read data. KILDA (Molitor & others, 2025) estimates KIV-2 diploid copy number directly from short-read FASTQ files using an alignment-free k-mer-counting approach

normalized against non-repetitive sequences. The proprietary Illumina DRAGEN LPA Caller similarly estimates KIV-2 copy number but requires the FPGA-accelerated DRAGEN platform. `vntr-calling-nf` (genepi, 2024) provides a Nextflow workflow for VNTR analysis but focuses on coding variant detection rather than large-VNTR copy number quantification and has primarily been validated using whole-exome sequencing data. Collectively, these tools do not provide an end-to-end solution for estimating haplotype-resolved copy number of large VNTRs from short-read whole-genome sequencing data at population scale.

Software design

GRiD integrates approaches from prior work for estimating large VNTR copy number from short-read sequencing (Hujoel et al., 2026; Mukamel et al., 2021). It provides a configurable, end-to-end workflow for estimating both diploid and haplotype-resolved large VNTR copy number from short-read whole-genome sequencing data at cohort scale. The pipeline is designed to be accessible to researchers without extensive bioinformatics expertise.

GRiD is implemented in Python (≥ 3.8) and follows a modular architecture organized into seven sequential pipeline steps, each implemented as an independent utility module under `grid/utis/`. The pipeline is orchestrated by `grid/pipeline.py` and driven by a user-supplied YAML configuration file, which specifies file paths, genomic coordinates, per-step parameters, and which steps to execute. The command-line interface is implemented with Click (Ronacher & others, 2024) and provides an entry point: `grid wgs` for whole-genome sequencing data.

The Whole Genome Sequencing pipeline proceeds as follows:

1. **Index verification / creation:** Ensures CRAM/BAM index files (`.crai/.bai`) exist for all input samples, creating them with `samtools` (Li et al., 2009) if needed.
2. **Read counting:** Counts properly paired reads in the target VNTR region using `pysam` (Heger & others, 2024), filtering by mapping quality and SAM flags.
3. **Coverage estimation:** Runs `mosdepth` (Brent S. Pedersen & Quinlan, 2018a) across the genome in binned mode to produce per-sample depth profiles.
4. **Coverage normalization:** Normalizes coverage within individuals (by sample mean depth) and across individuals (by a z-score transformation), then filters to high-variance regions enriched for VNTR signal.
5. **Nearest-neighbor identification:** Uses `scikit-learn` (Pedregosa et al., 2011) to compute Euclidean distances in normalized depth space and identify the top-N genomically similar neighbors per individual, providing a cohort-matched reference for CNV normalization.
6. **Diploid copy number estimation:** Computes per-exon diploid copy number for each individual by normalizing their read counts relative to their nearest neighbors' read counts, accounting for sample-specific sequencing depth.
7. **Haplotype inference:** Applies an IBD-based iterative phasing algorithm (Hujoel et al., 2026) that uses haplotype-matched neighbors to decompose diploid copy number estimates into haplotype-specific contributions.

All multi-sample steps are parallelized using Python's `concurrent.futures.ThreadPoolExecutor`, and progress is reported via the Rich terminal library (Willison, 2024). Configuration validation at pipeline startup provides early, informative error messages before any compute-intensive steps run.

Research impact statement

GRiD has been applied to >50,000 study participants with whole-genome sequencing data from multiple TOPMed cohorts and 1000G cohorts, comprising tens of thousands of participants across African, European, Hispanic/Latino, and admixed American ancestry groups. These

analyses form the basis of ongoing research manuscripts and demonstrate the applicability of GRiD to population-scale analyses and heterogeneous sequencing datasets.

GRiD provides a reproducible, open-source implementation of a previously custom analysis workflow, enabling large-VNTR copy number estimation from existing short-read whole-genome sequencing datasets. By packaging the complete workflow with documentation, example datasets, and automated installation, GRiD reduces the technical barriers to incorporating large VNTR analyses into population-scale genomic studies.

Mathematics

The mathematical framework implemented in GRiD follows previously published methods. Coverage estimation, normalization, and diploid copy-number estimation follow Mukamel et al. (Mukamel et al., 2021), while haplotype-specific copy-number inference follows Hujoel et al. (Hujoel et al., 2026). The equations below describe their implementation within GRiD; detailed methodological derivations and justification are provided in the original publications.

Coverage estimation. Following Mukamel et al. (Mukamel et al., 2021, Supplemental Materials, Section 2), weighted mean depth across the VNTR region from mosdepth (Brent S. Pedersen & Quinlan, 2018b) bins is computed as:

$$\text{Coverage} = \left\lceil 100 \cdot \frac{\sum_{i \in R} (\bar{C}_i \cdot (\min(e, r_{i,e}) - \max(s, r_{i,s})))}{\sum_{i \in R} (\min(e, r_{i,e}) - \max(s, r_{i,s}))} \right\rceil$$

where R is the set of bins overlapping the region $[s, e]$, $\bar{C}_i$ is the mean depth of bin i , and $r_{i,s}$ and $r_{i,e}$ are the bin boundaries.

Normalization. Following Mukamel et al. (Mukamel et al., 2021, Supplemental Materials, Section 2), let $D \in \mathbb{R}^{N \times M}$ be the depth matrix for N individuals across M genomic bins. Within-individual normalization gives $D_{ij}^{(1)} = D_{ij} / \bar{D}_i$, where $\bar{D}_i$ is the individual mean. Across-individual normalization then yields:

$$D_{ij}^{\text{norm}} = \frac{D_{ij}^{(1)} - \mu_j}{\sqrt{\mu_j}}$$

where μ_j is the population mean of $D_j^{(1)}$. Regions are filtered by a variance ratio σ_j^2 / μ_j , and the top fraction of high-variance regions are retained for neighbor computation.

Neighbor-normalized diploid copy number. GRiD uses the coverage-profile normalization approach described by Mukamel et al. (Mukamel et al., 2021, Supplementary Text, Section 2). For each individual, VNTR read depth is normalized relative to a set of samples with similar genome-wide coverage profiles. GRiD identifies these reference samples using nearest-neighbor search in the normalized depth space, implemented with scikit-learn (Pedregosa et al., 2011). The resulting diploid copy-number estimate for individual i is:

$$\widehat{\text{dipCN}}_i = \frac{r_i / s_i}{\frac{1}{|N_i|} \sum_{j \in N_i} r_j / s_j}$$

where N_i is the set of nearest-neighbor samples for individual i , and r_i / s_i represents the individual's VNTR read count normalized by sequencing depth.

Haplotype inference. Haplotype-specific copy-number inference follows the iterative phasing approach described by Hujoel et al. (Hujoel et al., 2026, Supplemental Materials, Section 5.3). In this framework, diploid VNTR copy-number estimates are partitioned between an individual's

two haplotypes using information from IBD-matched haplotypes, with haplotype-specific estimates updated iteratively. GRiD formalizes this procedure as:

$$h_{i,k}^{(t+1)} = \widehat{\text{dipCN}}_i \cdot \frac{\bar{h}_{N_{i,k}}^{(t)}}{\bar{h}_{N_{i,1}}^{(t)} + \bar{h}_{N_{i,2}}^{(t)}}$$

where $h_{i,k}^{(t)}$ is the estimated copy number of haplotype k for individual i at iteration t , and $\bar{h}_{N_{i,k}}^{(t)}$ is the mean haplotype copy number among IBD-matched reference haplotypes on haplotype k . At each iteration, the relative copy numbers of the IBD-matched haplotypes determine how the individual's diploid estimate is allocated between its two haplotypes. The estimates are then updated iteratively until convergence.

AI Usage Disclosure

The authors used GitHub Copilot and Anthropic Claude (Claude Sonnet 4) during the development of this software and the preparation of this manuscript. GitHub Copilot was used as an interactive coding assistant to suggest code completions, refactoring ideas, and boilerplate implementations. Claude was used to assist with code review, debugging, documentation development, and editorial improvements to the manuscript, including suggestions for clarity, organization, and grammar. AI tools were not used to generate scientific conclusions or interpret results. All AI-assisted code and manuscript edits were reviewed, validated, and, where necessary, modified by the authors before inclusion. The authors take full responsibility for the accuracy, correctness, and content of the software and manuscript.

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